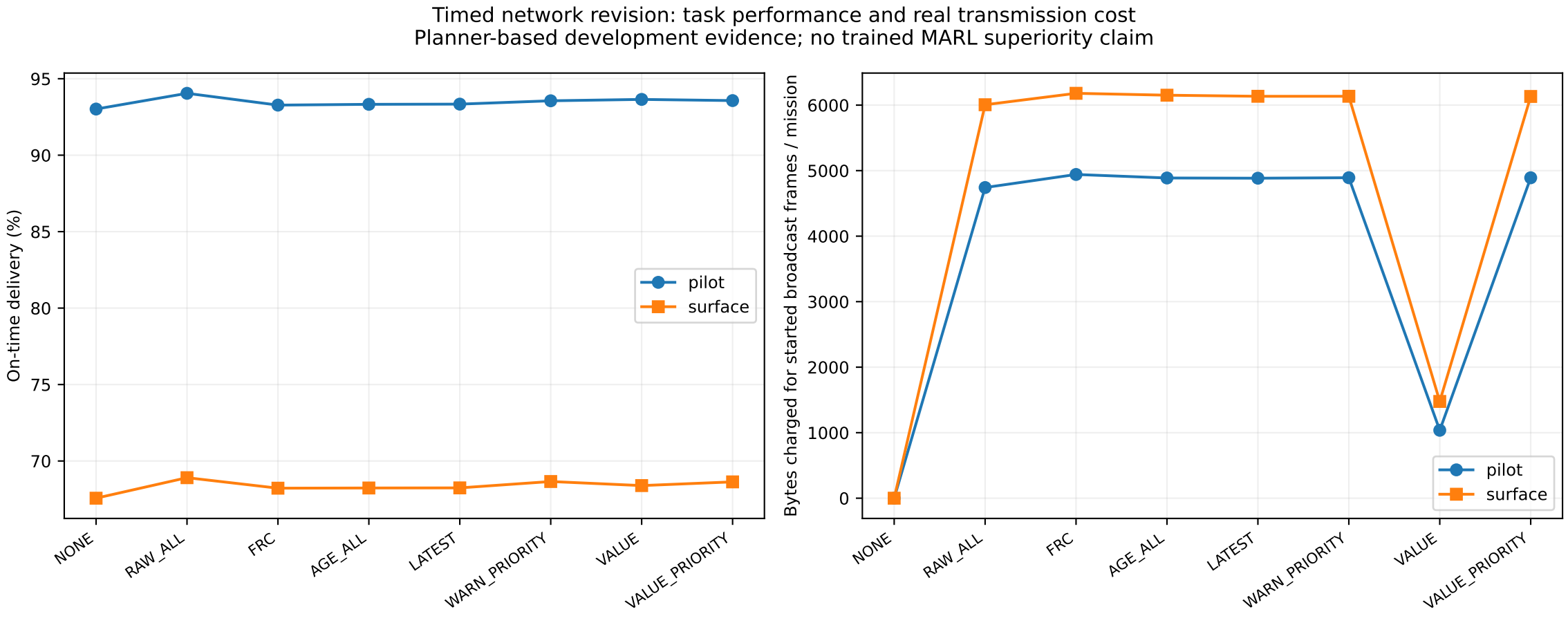
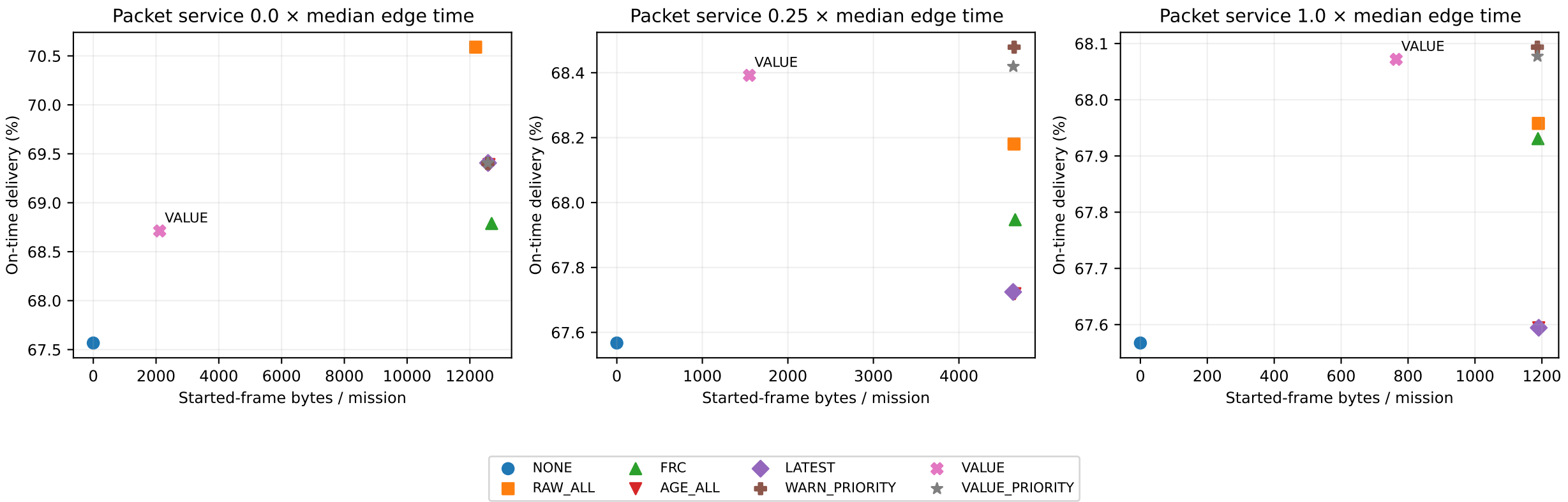
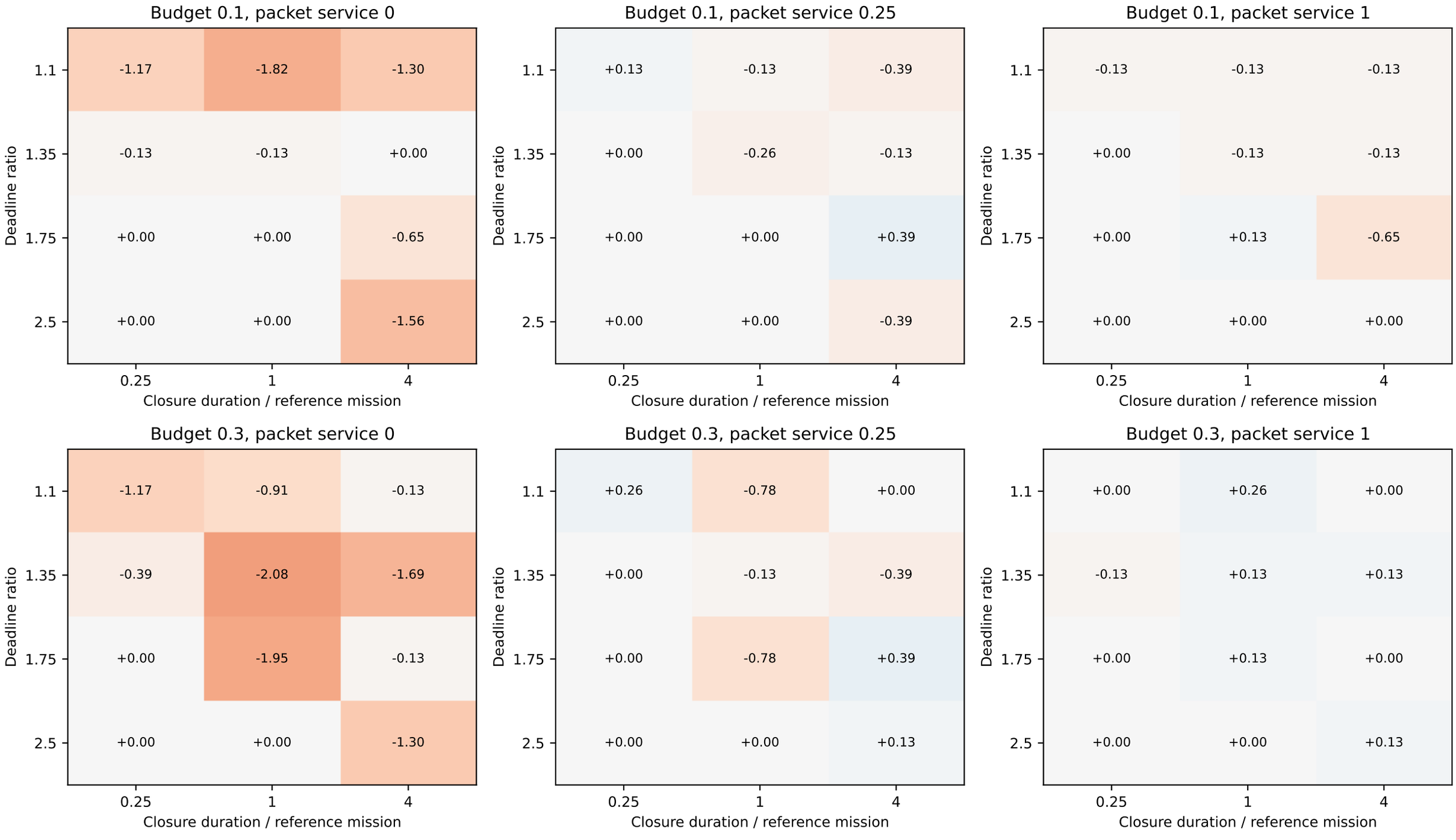
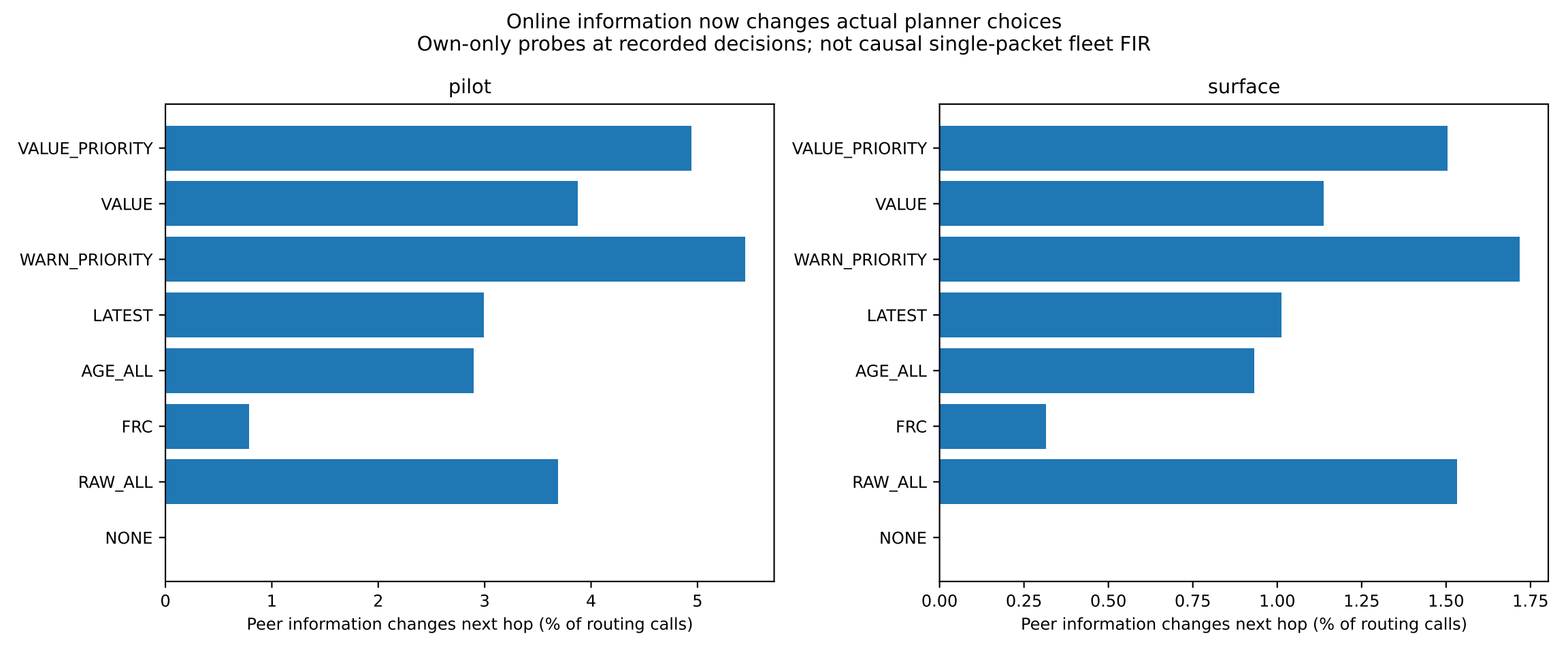
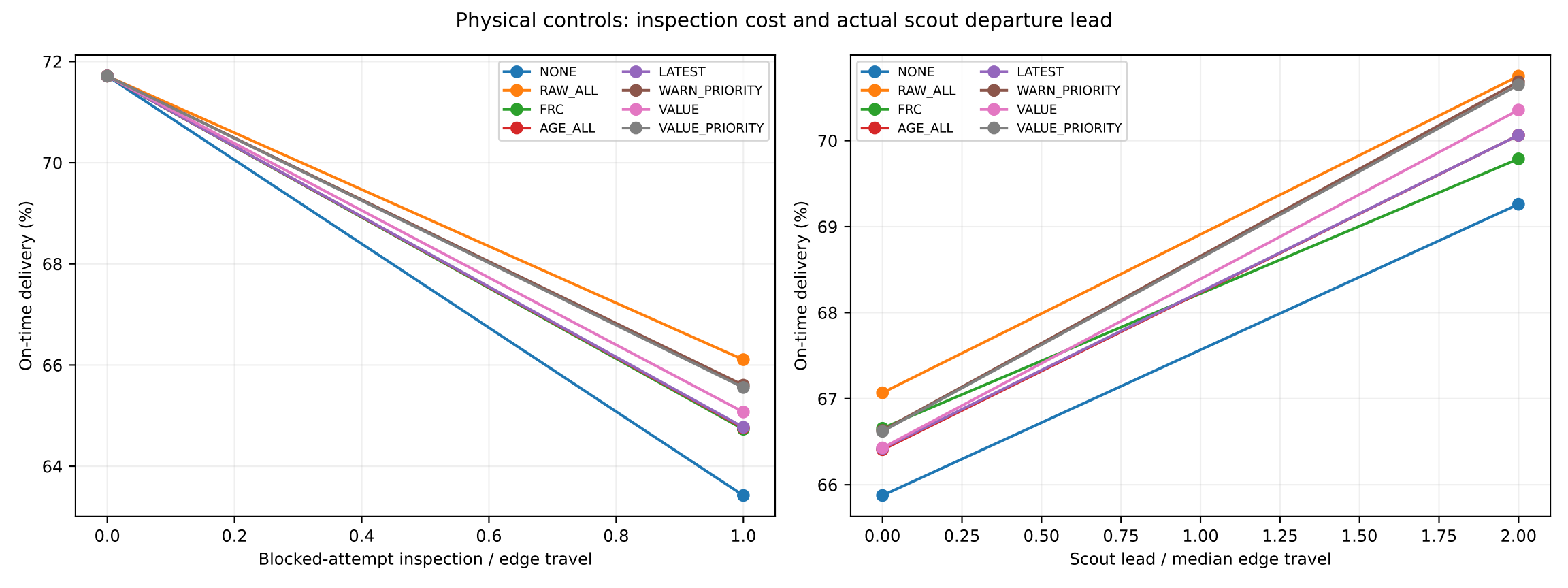
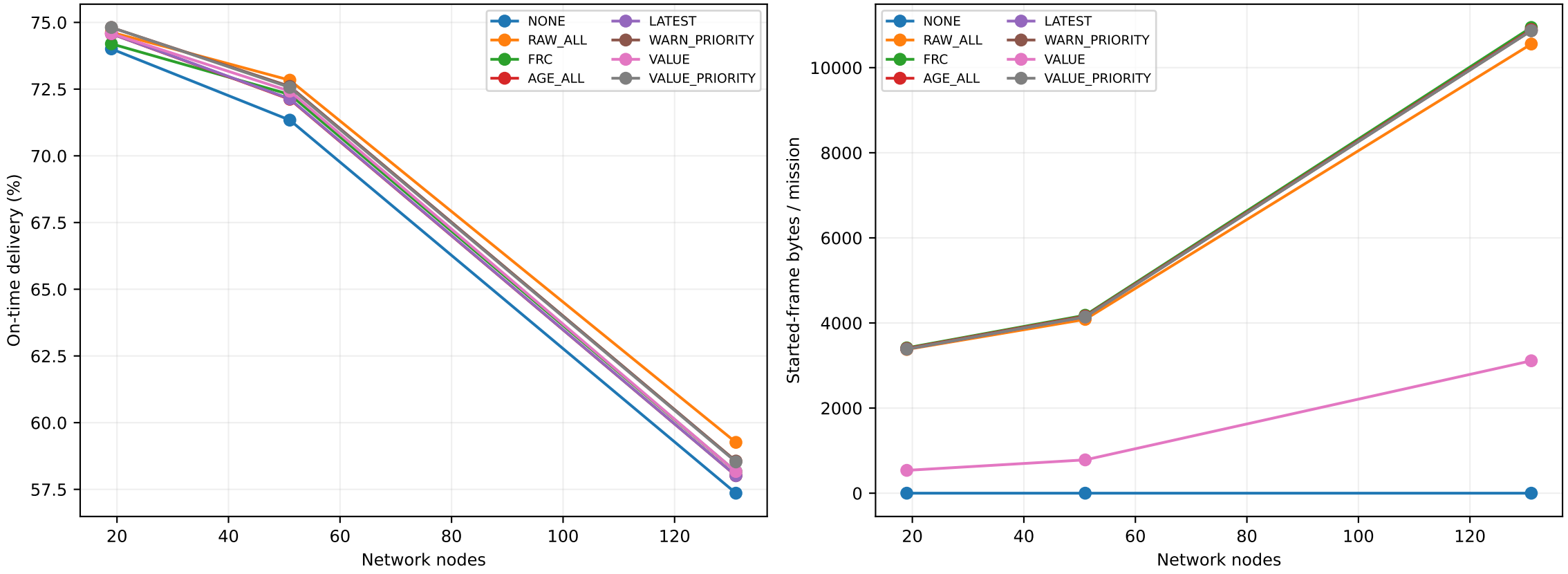

Timed network revision: task performance and real transmission cost
Planner-based development evidence; no trained MARL superiority claim
On-time delivery (%)
95
90
85
80
75
70
pilot
surface
NONE
RAW_ALL
FRC
AGE_ALL
LATEST
WARN_PRIORITY
VALUE
VALUE_PRIORITY
Bytes charged for started broadcast frames / mission
6000
5000
4000
3000
2000
1000
0
pilot
surface
NONE
RAW_ALL
FRC
AGE_ALL
LATEST
WARN_PRIORITY
VALUE
VALUE_PRIORITY

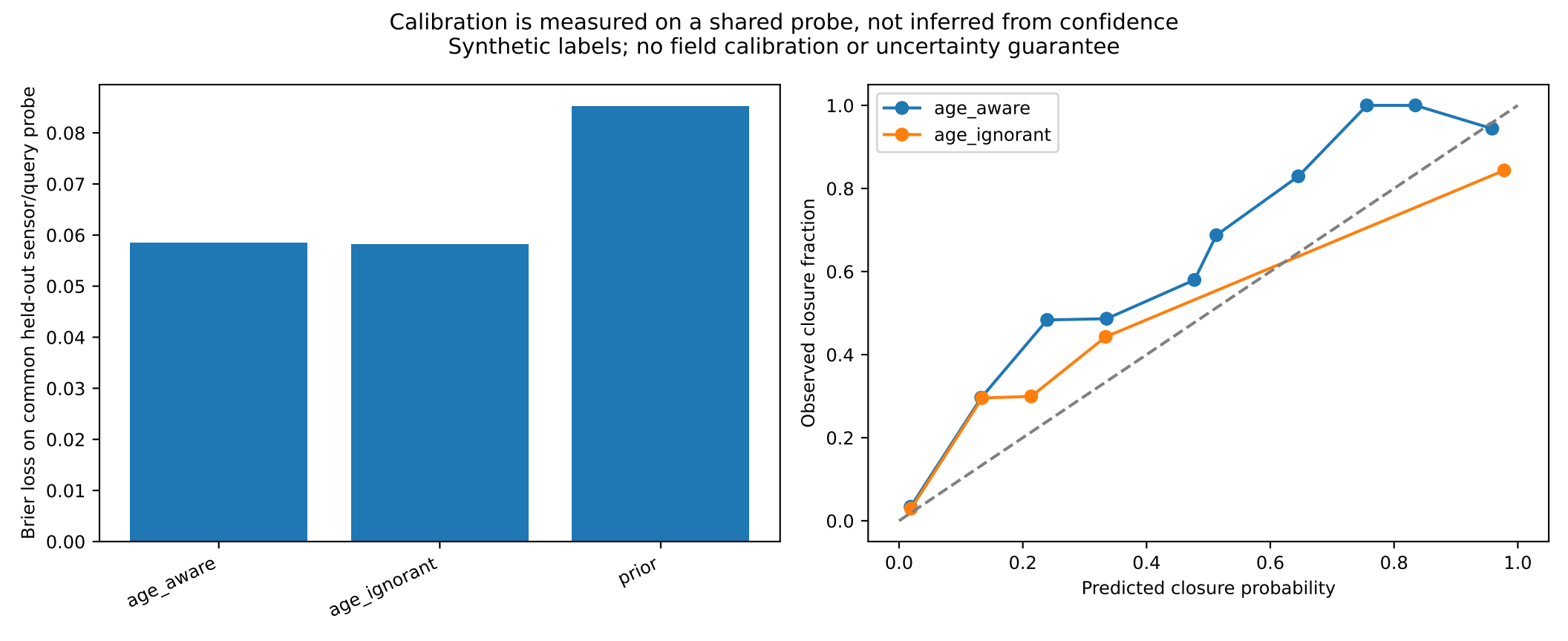

Whole declared surface: efficiency tradeoffs at each channel capacity
Packet service 0.0 × median edge time
Packet service 0.25 × median edge time
Packet service 1.0 × median edge time
On-time delivery (%)
Started-frame bytes / mission
VALUE
NONE
RAW_ALL
FRC
AGE_ALL
LATEST
WARN_PRIORITY
VALUE
VALUE_PRIORITY

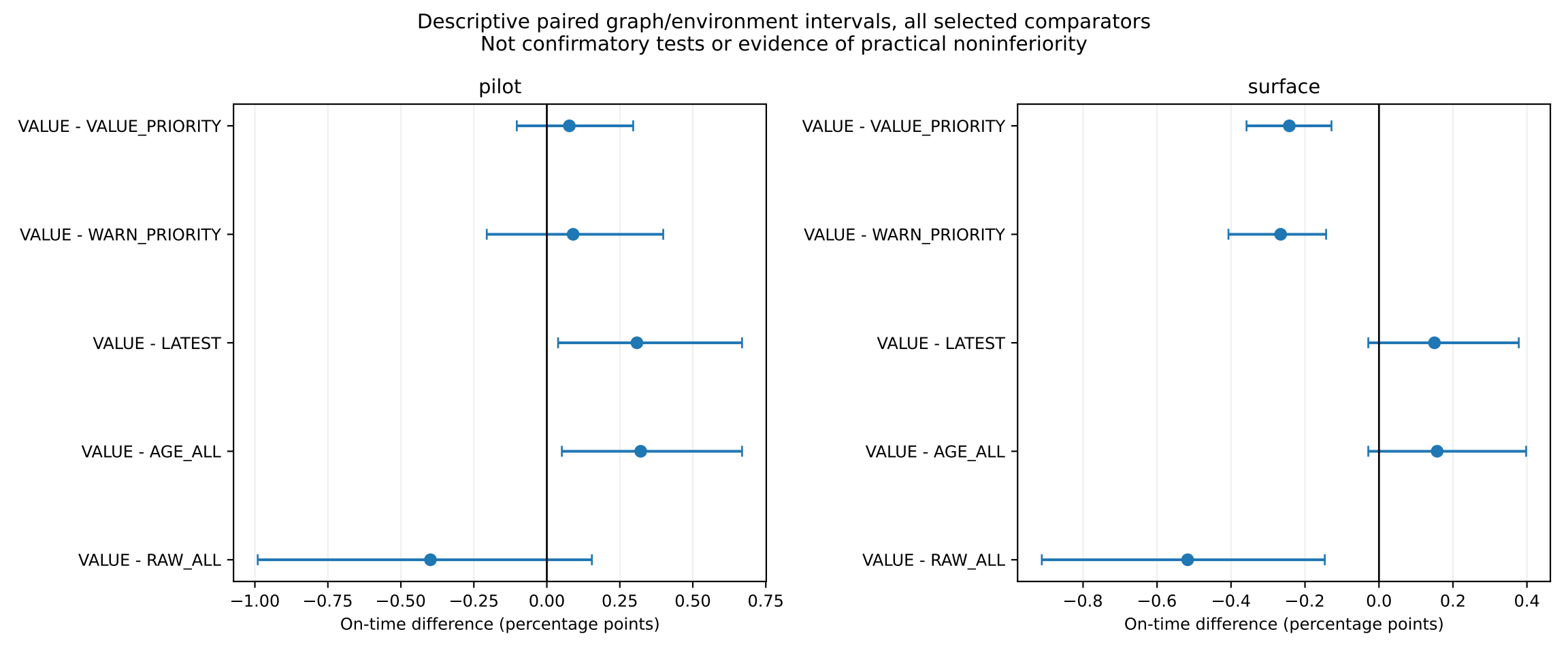

VALUE minus warning-priority on-time delivery (percentage points)
All deadline/persistence cells retained; averaged over sizes, leads and inspection factors
Budget 0.1, packet service 0
-1.17
-1.82
-1.30
-0.13
-0.13
+0.00
+0.00
+0.00
-0.65
+0.00
+0.00
-1.56
Budget 0.1, packet service 0.25
+0.13
-0.13
-0.39
+0.00
-0.26
-0.13
+0.00
+0.00
+0.39
+0.00
+0.00
-0.39
Budget 0.1, packet service 1
-0.13
-0.13
-0.13
+0.00
-0.13
-0.13
+0.00
+0.13
-0.65
+0.00
+0.00
+0.00
Budget 0.3, packet service 0
-1.17
-0.91
-0.13
-0.39
-2.08
-1.69
+0.00
-1.95
-0.13
+0.00
+0.00
-1.30
Budget 0.3, packet service 0.25
+0.26
-0.78
+0.00
+0.00
-0.13
-0.39
+0.00
-0.78
+0.39
+0.00
+0.00
+0.13
Budget 0.3, packet service 1
+0.00
+0.26
+0.00
-0.13
+0.13
+0.13
+0.00
+0.13
+0.00
+0.00
+0.00
+0.13
Deadline ratio
1.1
1.35
1.75
2.5
0.25
1
4
Closure duration / reference mission


Online information now changes actual planner choices
Own-only probes at recorded decisions; not causal single-packet fleet FIR
pilot
surface
VALUE_PRIORITY
VALUE
WARN_PRIORITY
LATEST
AGE_ALL
FRC
RAW_ALL
NONE
0
1
2
3
4
5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Peer information changes next hop (% of routing calls)
Peer information changes next hop (% of routing calls)


Physical controls: inspection cost and actual scout departure lead
On-time delivery (%)
Blocked-attempt inspection / edge travel
Scout lead / median edge travel
NONE
RAW_ALL
FRC
AGE_ALL
LATEST
WARN_PRIORITY
VALUE
VALUE_PRIORITY


Same model and fitted interpreter across 19/51/131 nodes
Two graph instances per size; this is not learned-policy scale convergence
On-time delivery (%)
Network nodes
Started-frame bytes / mission
Network nodes
NONE
RAW_ALL
FRC
AGE_ALL
LATEST
WARN_PRIORITY
VALUE
VALUE_PRIORITY


Calibration is measured on a shared probe, not inferred from confidence
Synthetic labels; no field calibration or uncertainty guarantee
Brier loss on common held-out sensor/query probe
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
age_aware
age_ignorant
prior
Observed closure fraction
Predicted closure probability
0.0
0.2
0.4
0.6
0.8
1.0
age_aware
age_ignorant


Descriptive paired graph/environment intervals, all selected comparators
Not confirmatory tests or evidence of practical noninferiority
pilot
surface
VALUE - VALUE_PRIORITY
VALUE - WARN_PRIORITY
VALUE - LATEST
VALUE - AGE_ALL
VALUE - RAW_ALL
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
On-time difference (percentage points)
On-time difference (percentage points)